

## MediExtract: business and technical trade-off analysis

Youtube Video: <https://youtu.be/EtTjQZDcFD0>

Every figure here is reproducible from an artefact in this repository; `tests/test_guardrails_doc.py` fails if a quoted metric drifts from the file it came from.

**The decision being made.** Dr. Aisha, a general practitioner in a Singapore polyclinic, dictates a consult in ninety seconds, then retypes medication, dose and frequency into the EHR to close the chart. Ambient scribes already produce fluent prose; none show her *where* a value came from, and she will not sign an AI-generated dose she cannot verify in under three seconds. Every trade-off below follows from optimising verification speed over fluency.

### Technical trade-offs

| Decision      | Chosen                              | Rejected                | The evidence that decided it                                                                               |
|---------------|-------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------|
| Extractor     | Foundation model, structured output | Rules/lookup as primary | 0.730 pooled recall against <b>0.381</b> for a committed regex + 14,689-name RxNorm baseline               |
| Gates         | Deterministic Python                | Model self-check        | A wrong dose is expensive; if statements are auditable and free                                            |
| Retrieval     | None                                | RAG                     | The answer is always inside the note. Retrieval adds a failure mode (wrong chunk) for zero possible recall |
| Orchestration | One call, no tools                  | Multi-agent             | Changes the <i>threat model</i> , not just the bill                                                        |
| Framework     | None, one file                      | LangChain               | Every request parameter auditable; cost model verified to <b>0.005%</b> drift                              |
| Hosting       | Rented API                          | Self-hosted Qwen        | Reasoned, not measured. Crossover $\approx$ <b>1,700 physicians</b>                                        |

**Why rejecting agents is load-bearing.** OWASP's component-versus-actor boundary means the LLM Top 10 (2026) applies to a single-pass extractor and the Agentic Top 10 does not. Adding one tool would change what the system *is*, invalidating the entire security specification. That is a governance saving, not a latency one.

**The gates are the product.** A field that cannot be grounded is wiped to null never to a display string. Precision 0.750, silent-failure rate 0.252 **31 of those 36 fields are one defect, drug selection in a multi-drug note, owned by the schema and reachable by no gate**; two controls proposed to reduce it were measured and refused. Enforcing

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verbatim evidence spans roughly doubles output tokens, about **\$1.90 per physician-year**: the explicit cash price of the safety property.

### Business trade-offs

| Line                          | Measured                    |
|-------------------------------|-----------------------------|
| Cost per note                 | <b>\$0.000617</b>           |
| Cost per <i>correct</i> field | <b>\$0.000383</b>           |
| Per physician-year            | <b>\$3.58</b>               |
| Median latency                | 1,293 ms (66/67 inside 3 s) |

**Accuracy beats price by three orders of magnitude.** At SGD 100/h clinician time, five percentage points of accuracy are worth **561×** a token-price saving. Token price is therefore not a decision variable in this workflow which inverts the usual build-vs-buy instinct. Self-hosting to save inference cost would optimise the wrong term.

**The highest-value change is a schema change, not a model change.** Where labeller and model name the same drug, dose is correct 24/26 and frequency 30/30 the residual error is almost entirely the single medication slot. Moving to a medication *list* is projected to save ~**\$3,600 per physician-year** in clinician time against a **\$3.58** annual inference bill.

### The trade-offs that cost us something

#### Structural dependency:

Stripping ALL-CAPS section headers drops allergy recall **0.900 → 0.650** (McNemar  $p = 0.0625$ , the exact floor at  $n = 20$ ). Enumerating all thirty gates showed why none fired: **every gate is a precision control; none is a recall control**. Re-dictating the same consults around byte-identical labels moved recall  $0.730 \rightarrow 0.750$  ( $p = 0.6072$ , a null), localising the dependency to the header specifically rather than to formatting in general.

**An LLM judge was built and refused:** The adoption gate was written before the code: 0.9 agreement on *both* classes. Measured 0.9467 where gold says correct, **0.2093** where gold says wrong and a judge answering "correct" to everything scores 0.8396 against the real judge's 0.8284. It does not beat a rubber stamp. Automated evaluation was rejected; hand labelling still stands behind every prompt change. Cost of learning this: \$0.0696.

**Prompt-level defence is not a control:** A live red-team run (\$0.0173 of a \$0.50 allocation) found the model **obeyed 5 of 10 prompt injections**, planting warfarin 10 mg daily and 5000 mg into fields. The deterministic tripwire downgraded all five to review, so **none reached a payload as verified**. With the encoding pre-filter disabled, 3 of 16 homoglyph vectors put a lookalike (metformin, Cyrillic e) into a field marked found the grounding gates cannot catch these, because a homoglyph in the note *is* verbatim in the note. Defence in depth, measured: the probabilistic layer failed, the deterministic layer held.

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### What was given up:

- **Recall for precision.** Abstention is the default on anything ungroundable. A blank costs a manual entry; a wrong dose costs more.
- **Breadth for auditability.** One slot per field, not a list carrying most of the residual error, kept because changing it invalidates the sealed labels.
- **Automation for trust.** No judge, no self-grading; every score traces to a hand-labelled answer key.
- **Statistical power for honesty.**  $n = 67$ . Wilson intervals on every rate, a paired test on every delta, and a large p-value reported as *the evaluation is too small to tell* rather than as equivalence.

### Verification

519 tests pass standard and under python -O, which strips assert the reason no runtime gate may be one. tests/test\_guardrails\_doc.py parses the security specification with Python's ast module and fails if a quoted metric no longer equals the artefact it names. Five corpora are hash-sealed at 0444 and tagged.

Total project spend: **\$0.32 of an \$8 ceiling.**